nature methods

Article

https://doi.org/10.1038/s41592-025-02899-6

Novae: a graph-based foundation model for spatial transcriptomics data

Received: 24 September 2024

Accepted: 10 October 2025

Published online: 10 December 2025

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Spatial transcriptomics is advancing molecular biology by providing high-resolution insights into gene expression within the spatial context of tissues. This context is essential for identifying spatial domains, enabling the understanding of microenvironment organizations and their implications for tissue function and disease progression. To improve current model limitations on multiple slides, we have designed Novae (https://github.com/MICS-Lab/novae), a graph-based foundation model that extracts representations of cells within their spatial contexts. Our model was trained on a large dataset of nearly 30 million cells across 18 tissues, allowing Novae to perform zero-shot domain inference across multiple gene panels, tissues and technologies. Unlike other models, it also natively corrects batch effects and constructs a nested hierarchy of spatial domains. Furthermore, Novae supports various downstream tasks, including spatially variable gene or pathway analysis and spatial domain trajectory analysis. Overall, Novae provides a robust and versatile tool for advancing spatial transcriptomics and its applications in biomedical research.

Spatial transcriptomics[1,2] data provide invaluable insights into cellular interactions within their microenvironment and the complexities of tissue organization. A key advantage over current single-cell RNA sequencing (scRNA-seq)[3] is that spatial transcriptomics maintains the spatial context of cells, enabling a deeper understanding of how cells interact within their native environments. Technologies in spatial transcriptomics can be broadly categorized into two types: (1) next-generation sequencing[4] (NGS)-based methods that offer whole transcriptome sequencing, and (2) imaging-based techniques such as Xenium[5], MERSCOPE[6] or CosMX[7], which provide subcellular resolution. The former allows for comprehensive gene analysis but lacks fine spatial detail, whereas the latter offers detailed spatial resolution but with a limited gene panel size. As imaging-based technologies continue to evolve, they expand their gene panel capabilities, enabling the inclusion of larger panels or the replacement of low-quality genes during studies; however, this flexibility often results in experiments conducted across different machines or using varying panels, which introduces new challenges. In more general cases, when performing analysis over multiple spatial transcriptomics slides (for both NGS or imaging-based techniques), it is common to observe a strong batch effect, such that it can be challenging to identify common spatial patterns across multiple slides without careful consideration of the batch effect.

A key focus in spatial transcriptomics is the identification and categorization of spatial microenvironments, often referred to as spatial domains or niches. Various methods, such as STAGATE[8], GraphST[9], SpaceFlow[10] and SEDR[11], have been developed for this purpose. While these methods show promising results (especially with NGS-technologies with spot resolution like Visium), they are limited by (1) their reliance on predefined gene panels; (2) sensitivity to batch effects; and (3) dependence on external tools like Harmony[12] for batch-effect correction and Leiden[13] or Mclust[14] for clustering. These dependencies can slow down processing and reduce robustness, as

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https://doi.org/10.1038/s41592-025-02899-6

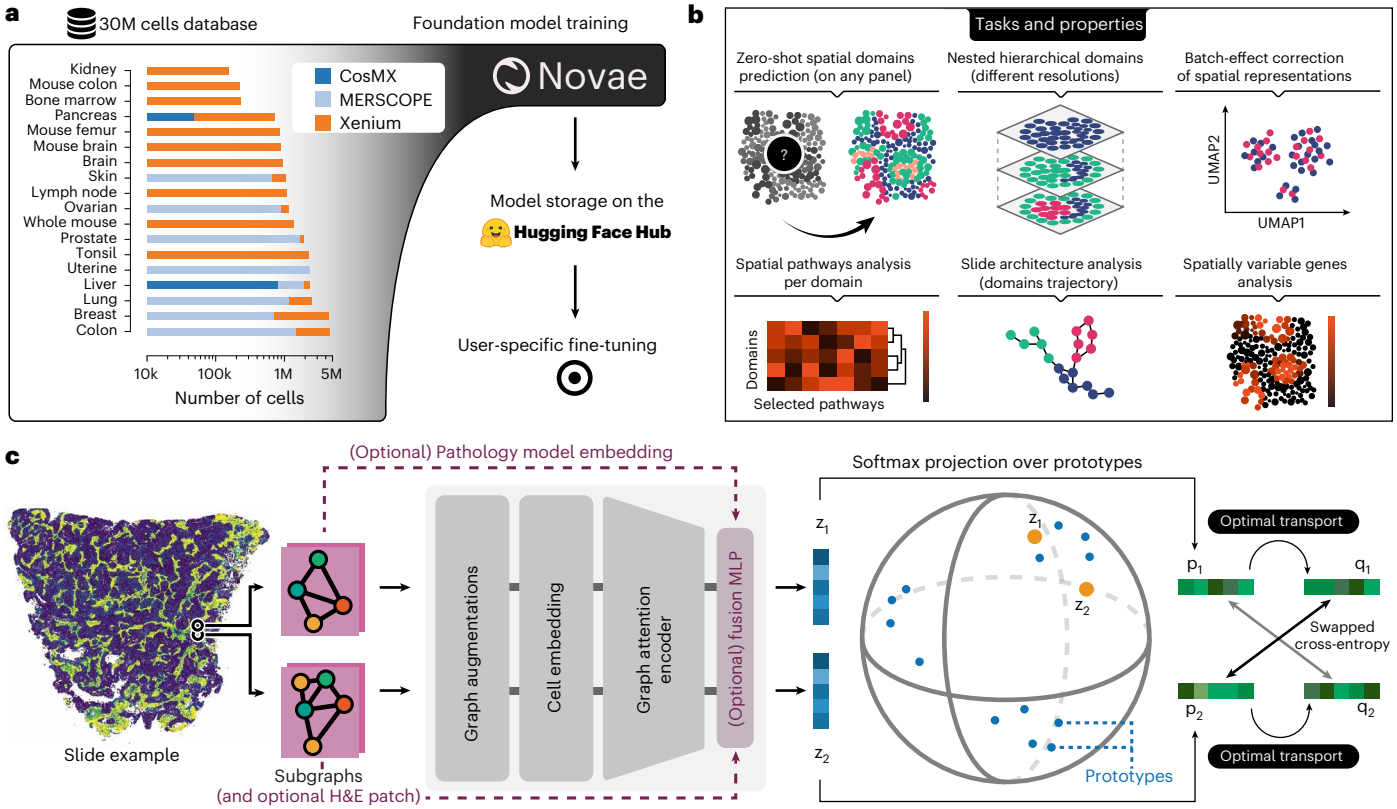


Fig. 1 | Overview of Novae. a, The dataset is composed of 29 million (M) cells across 18 different tissues for three different single-cell spatial transcriptomics technologies. This dataset was used to train Novae, which was shared online on the Hugging Face Hub. Users can easily load the pretrained model for reuse or fine-tuning. The Hugging Face Hub logo is reproduced from their open-source GitHub repository (https://github.com/huggingface/huggingface_hub). **b**, A visual summary of the key tasks and capabilities of Novae. Notably, Novae can infer spatial domains in a zero-shot manner, organizing them hierarchically. Additionally, based on learned representations, Novae can align representations across different batches. Other downstream tasks include spatial pathway analysis, slide architecture analysis and SVG analysis. **c**, An overview of Novae's underlying method. The process involves extracting two nearby subgraphs of cells (or spots), which are then augmented and embedded. These graph inputs are processed through a graph attention encoder, generating representations for both subgraphs. These representations are projected onto prototypes, learnable vectors from the latent space. Finally, an optimal transport task is applied to the assignment probabilities over these prototypes, and a swapped cross-entropy loss is computed for back propagation. When an aligned H&E image is available, we extract an H&E patch corresponding to each subgraph. From each patch, we compute an embedding using a pretrained pathology foundation model, which is then fused with the graph representation using a multilayer perceptron (MLP).

external tools need to be re-run for each new analysis or when adjusting spatial domain resolutions (choosing different number of spatial domains). Additionally, owing to their reliance on specific gene sets, these methods often necessitate training on the intersection of gene sets, which can significantly reduce the number of available genes and, consequently, impact performance. Notably, even when applied to slides with a shared panel, these models tend to identify primarily slide-specific domains, which limits the comparison of domains across a broader study and reduces the potential for discovering new spatial biomarkers.

To address these limitations, we introduce Novae, a self-supervised[15] graph attention network[16] that encodes local environments into spatial representations. Unlike existing methods, the same Novae model can operate with multiple gene panels, allowing for the application across diverse technologies and tissues. It includes native batch-effect correction methods, directly correcting for variations and enhancing robustness and scalability. Therefore, Novae's design allows it to seamlessly integrate data from different platforms and gene panels without compromising performance. We trained Novae on a large dataset comprising 78 slides, representing nearly 30 million cells across 18 tissues and three different subcellular resolution technologies (Xenium[5], MERSCOPE[6] and CosMX[7]). This broad training allows Novae to compute relevant representations via zero-shot[17] or fine-tuning on new slides from different tissues. These representations can be directly used for spatial domain identification, eliminating the need for external clustering tools. Beyond spatial domain identification, these representations can be applied to various downstream tasks, including (1) spatial domain trajectory analysis; (2) spatially variable gene (SVG) analysis; and (3) spatial pathways analysis. Novae's versatility, robustness and ease of use make it a powerful tool for advancing spatial transcriptomics research within the scientific community.

Results

A foundation model for spatial transcriptomics

Novae is a graph-based foundation model[18,19] that we trained on a large single-cell spatial transcriptomics dataset composed of about 30 million cells. This model, aware of 18 different tissues, is hosted on a public hub and can be reused by the user for multiple downstream tasks on any technology or gene panel, as illustrated in Fig. 1a. The main application of Novae is to learn spatial domains, which can be performed with the shared model without any retraining (called zero-shot inference[17]). If desired, the user can also retrain Novae to obtain refined results (fine-tuning). Two distinguishing properties of Novae are that (1) it provides a nested organization of spatial domains for different resolutions and (2) it natively corrects batch effect across slides. This consistent domain assignment across slides enables comparison analyses of a study containing multiple slides. Additionally, Novae can perform multiple downstream analysis tasks, such as (1) spatial pathway analysis;

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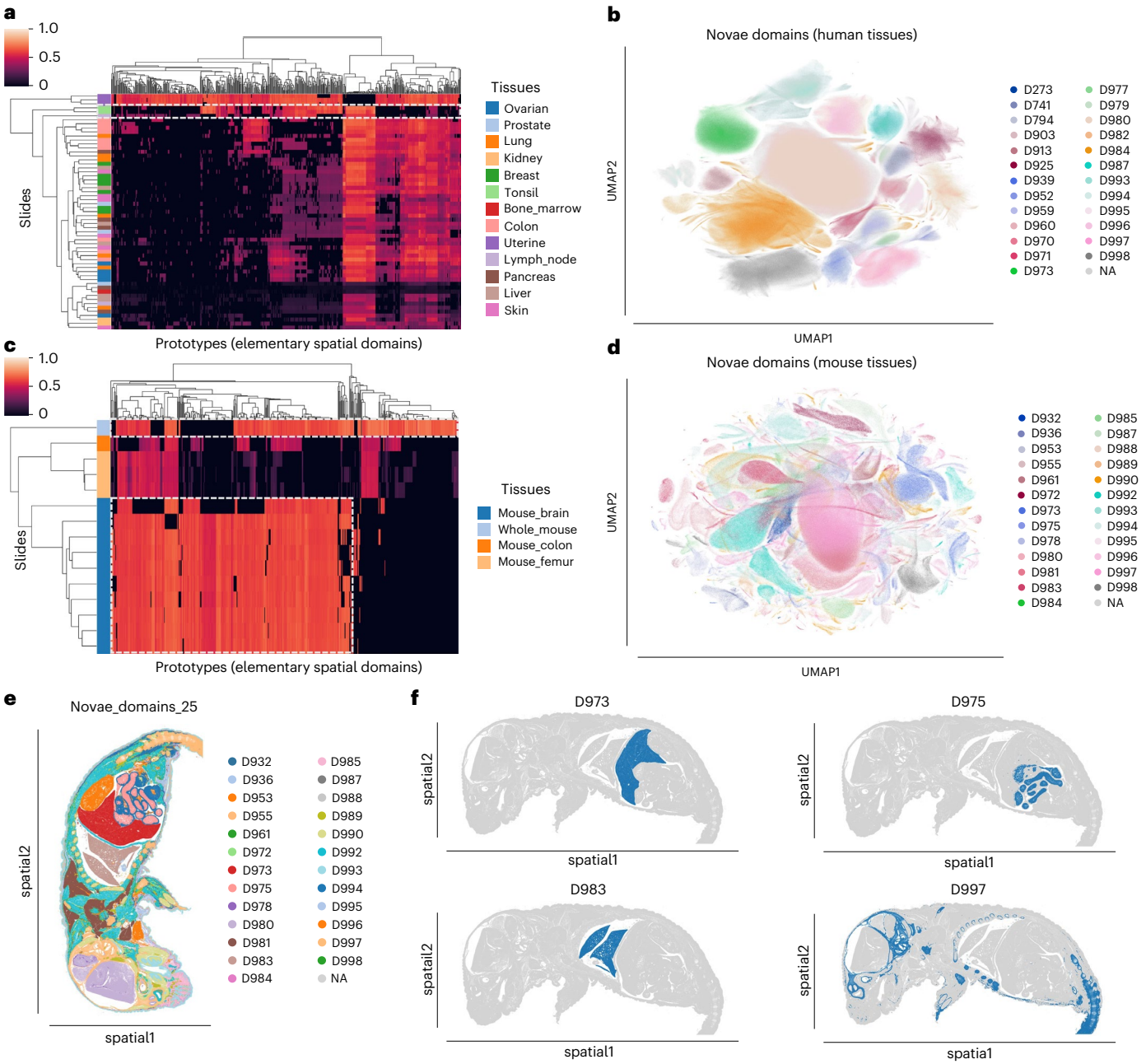


Fig. 2 | Spatial domains across tissues and species. a, Heatmap showing the spatial domain weights for each slide, organized by tissue type. A weight of 0 (black) indicates the absence of the domain in that slide, and higher weights represent increased confidence in the presence of the domain. Dashed lines show similar spatial domains distribution in tonsil and lymph node slides. **b**, UMAP visualization of spatial domain representations in human tissue slides. Each dot corresponds to the spatial context representation of a cell. Isolated cells (that is, cells without any neighbors in the adjacency graph) are labeled as NA (not applicable). **c**,**d**, Similar figures to **a** and **b**, but for the mouse samples. Dashed lines in **c** show that the mouse brain slides have a strong inter-slide similarity, while the whole mouse sample has slide-specific domains. **e**, Spatial domains identified for the whole mouse sample. **f**, Zoom-in on four domains in the whole mouse: liver (D973), intestine (D975) within the abdominal cavity, lung lobes (D983), bones and costal cartilage (D997).

(2) spatial domain organization/architecture analysis; or (3) spatial variable genes analysis. These tasks and properties are summarized in Fig. 1b. Written in Python, Novae is also integrated within the scverse[20] ecosystem and works natively with standard libraries such as AnnData[21], SpatialData[22] and Sopa[23].

In terms of technical details, Novae is a graph-based neural network that is trained in a self-supervised manner based on the SwAV[24] framework. It learns representations of the local microenvironment at the single-cell (or spot) resolution. More specifically, for each cell or spot, a representation of its neighborhood is provided by a Graph Attention Network[16], after embedding each cell into a panel-invariant representation. To effectively assign domains, we learn embeddings in the latent space, called prototypes, which represent elementary spatial domains. Depending on the desired level of resolution, these elementary domains are regrouped to form the domains at the desired resolution. We project the spatial domain representations into these prototypes to get probabilities of assignments. These probabilities are corrected via an algorithm of Optimal Transport[25] to ensure a smooth representation of the different prototypes. Afterwards, we use cross-entropy loss to make the representation of two cells in the same domain closer together. This methodological approach is illustrated in Fig. 1c.

Pan-tissue spatial domains

As outlined in the previous section, Novae can analyze multiple panels and tissues, with the ability to identify both shared and tissue-specific spatial domains. While we anticipate some overlap in domains across tissues, we also expect certain domains to be specific to particular conditions or diseases. In Fig. 2a,c, we present the spatial domains identified across various slides for both human (Fig. 2a) and mouse (Fig. 2c) tissues. On these figures, one row corresponds to one slide and is colored on the left by the corresponding tissue, and one column corresponds to one prototype. The two hierarchical clusterings show how the slides and prototypes are grouped. Notably, the lymph node and tonsil exhibited similar spatial domain distributions, with some domains also observed in other tissues such as breast and lung, whereas the two uterine slides can be seen as outliers. Also, the heatmap dendrogram regroups slides with similar patterns of domains, and is not a perfect tissue match. Indeed, the spatial domains also include context related to diseases or certain mechanisms, which can be retrieved from multiple different tissues. For instance, a lung tumor and a breast tumor may share some cancer-related domains that cannot be found in healthy breast tissue[26]. The Uniform Manifold Approximation and Projection (UMAP)[27] representation of spatial context is shown in Fig. 2b,d, where each dot corresponds to a cell's spatial context, clustered into distinct spatial domains. Regarding the mouse study, we considered brain slides, as well as colon, femur and also a whole mouse sample, as detailed in 'Datasets used'. More specifically, the mouse brain slides showed strong inter-slide similarity, as many domains were consistently identified across all slides (Fig. 2c). Additionally, we observed that many domains, including 'brain-specific' and 'colon-specific' domains, were retrieved in the whole mouse sample, but many prototypes are used exclusively for the whole mouse sample. Examples of spatial domains for the whole mouse sample are shown in Fig. 2e. For instance, we find the areas characterizing the bones and costal cartilage (D997), the lung lobes (D983), the liver (D973) and the intestine (D975) in the abdominal cavity[28], as highlighted in Fig. 2f. In summary, while the results shown in Fig. 2 were expected, it shows the capability of Novae to train across multiple tissues, while avoiding overcorrection (the identification of all domains in all slides) or under-correction (all spatial domains being specific to one slide).

Integration and continuity of the spatial domains

In this study, we compared Novae in both zero-shot and fine-tuning modes to six state-of-the-art methods: SpaceFlow[10], GraphST[9], SEDR[11], STAGATE[8], NicheCompass[29] and Scanpy[30] (more details are in 'Spatial domains assignment benchmark'). We evaluated the performance of these methods across three distinct test cases. In the first test case, we used the breast dataset, composed of two slides with different gene panels. As mentioned earlier, Novae can be trained across multiple gene panels; hence, a single Novae model was trained for both panels. In contrast, the other methods were trained on the intersection of 185 common genes across the two panels (as illustrated in Fig. 3a), followed by batch-effect correction using Harmony[12] and clustering with mclust[14] (using 7, 10 and 15 clusters). To evaluate model performance, we used the FIDE (F1-score of inter-domain edges) score and Jensen–Shannon Divergence (JSD) score to assess spatial domain continuity and cross-slide homogeneity, respectively (see 'Metrics for model comparison'). Figure 3b presents the results of this benchmark, highlighting a notable improvement in performance by Novae, even in the zero-shot case (using a pretrained model directly).

The second test case involved the colon dataset, where Novae was again trained across all slides. For the other methods, due to the limited intersection of genes between panels, we employed a different approach: a separate model was trained for each slide (as illustrated in Fig. 3c). The spatial representations from each model were then concatenated, followed by batch-effect correction and clustering. Figure 3d presents the result metrics for all methods on the colon dataset, again showing a superior performance by Novae in both zero-shot and fine-tuning modes.

For visual comparison, Fig. 3e shows the Novae spatial domain assignments for the breast samples (Supplementary Fig. 9 shows the results of the other methods and Supplementary Fig. 5 shows the results for the colon samples), showing that Novae can better identify shared domains across multiple distinct slides. Also, Fig. 3i shows that Novae better aligns the batches/slides in the latent space. Additionally, we show that Novae is not overcorrecting, as it identifies the domain D503 as being much more represented on the Xenium slide than the MERSCOPE slide (more biological details in Supplementary Fig. 2). A more detailed study regarding the overcorrection can be found in 'Robustness to missing domains and perturbations'. Regarding the biological interpretation of Novae's results, domains D498 and D499 were identified as stromal regions characterized by the expression of *FN1* and *COL1A1* (Supplementary Fig. 2), which are present in both samples. Domain D504 was composed of glandular cells, marked by the expression of *TAPBP*, *PGR* and *CDH1*. In the Xenium sample, an expansion of domain D503 was observed, potentially resulting from clonal expansion of cancer cells. This expanded D503 became embedded within the stromal areas of D498 and D499, while excluding the immune cell-rich region D485, which is known for high levels of *PTPRC*, *CD52* and *CD3E*[31]. This exclusion pattern, as depicted in the PAGA[32] graphs in Supplementary Fig. 2b,d, is characteristic of tumors with an immune-excluded phenotype[31].

The third comparison was conducted on a synthetic dataset consisting of five slides and seven spatial domains (see 'Datasets used' for more details). Unlike the previous cases, this dataset used the same gene panel across all slides, allowing the other methods to run without requiring gene panel intersection. In this case, we did not use Novae in zero-shot mode, as the gene expression was synthetic, meaning the generated cell types may not correspond to real data. Likewise, we did not use NicheCompass[29] here, as it relies on prior gene programs that reflect real biological interactions (such as ligand–receptor pairs and transcription factor–target relationships) to guide its embedding. Because these programs must map to actual genetic features, using purely synthetic data (with no direct biological meaning) breaks NicheCompass's fundamental assumptions about gene–gene interactions. The models were evaluated using the adjusted Rand index (ARI)[33] for clustering accuracy against the known ground truth, as well as the FIDE score. Figure 3f shows the ARI of the different methods across five seeds, and Fig. 3g presents the FIDE score across five seeds. Again, Novae outperformed the other methods, demonstrating higher ARI and FIDE scores with notably low s.d. in ARI (Fig. 3f). The ground truth and the spatial domain predictions of the different models are shown in Supplementary Fig. 4.

Time and memory efficiency

After running inference (computing the cell's spatial representations), Novae can perform the attribution of the niches and correct batch effect in a short time. Indeed, the attribution of spatial domains is a mapping between the prototypes and the desired resolution (see 'Assignment to spatial domains'), which is performed in a short time. Regarding batch-effect correction, as the categorical domain assignments are already corrected through Novae, we can use the assignments to align the spatial representations (as detailed in 'Batch-effect correction'). This vectorial operation is also fast and in linear time. In contrast, for the two latter operations, the other state-of-the-art models depend on external tools, usually (1) Harmony[12] for batch-effect correction and (2) Leiden[13] or mclust[14] to assign spatial domains to the representations. As shown by Fig. 3h (bottom), using Harmony and Leiden can be slow, especially on large datasets of millions of cells. Indeed, this can take up to several days on 6 million cells, whereas Novae can perform these two operations in several seconds. Furthermore, during experimentation, it is common to try multiple resolutions of spatial domains, hence

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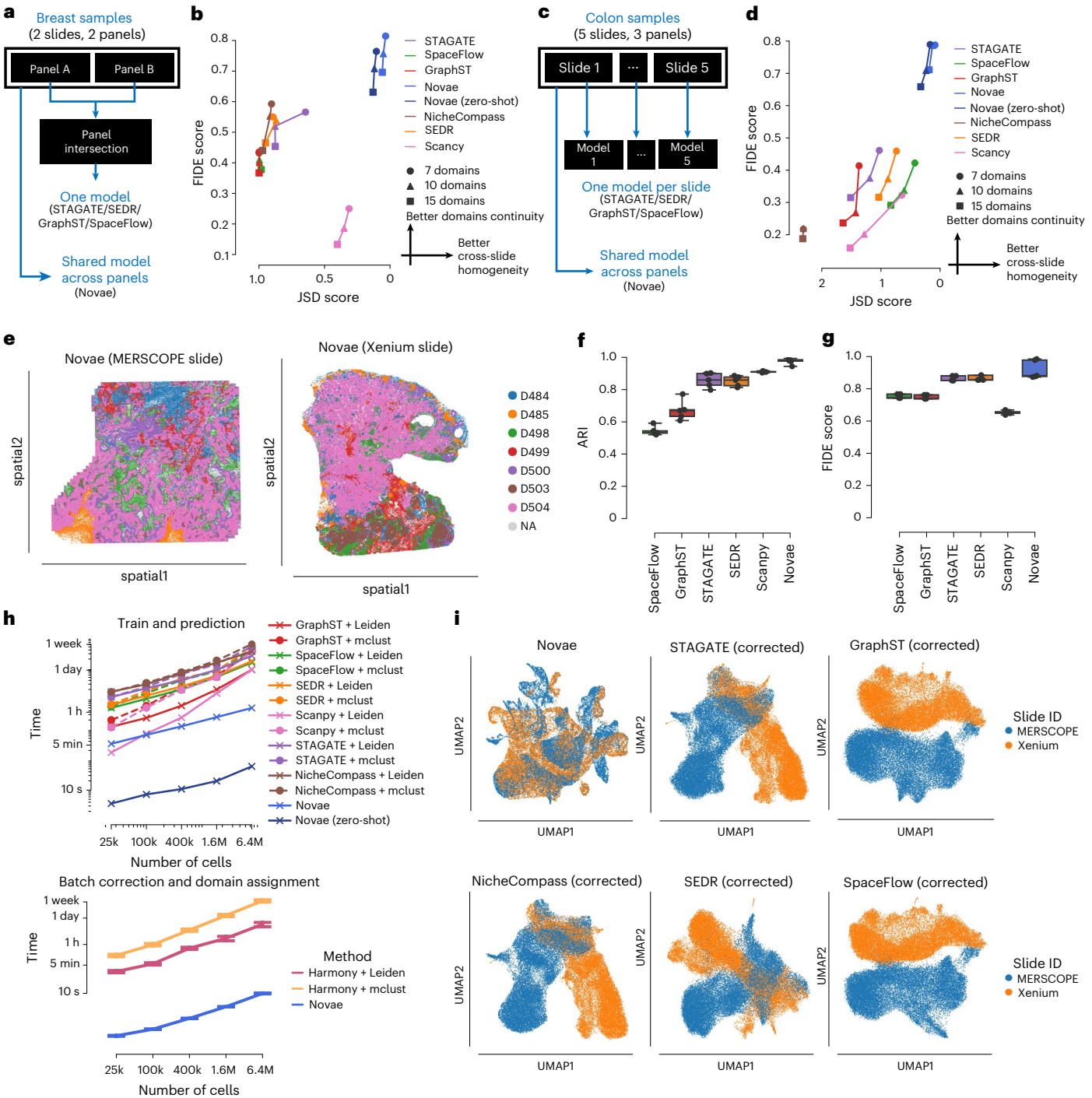


Fig. 3 | Benchmark of the integration and continuity of the spatial domains assignment. a, Schematic representation of the benchmarking approach for the breast dataset. The same Novae model can be trained on both panels, while other models were run using genes shared between the panels. **b**, FIDE and JSD scores of different methods for three domain counts (7, 10 and 15) on breast samples. The FIDE score evaluates domain continuity, whereas the JSD score assesses cross-slide homogeneity. **c**, Schematic representation of the benchmarking approach for the colon dataset. The same Novae model can be trained on all panels, whereas other models were run in a one-per-slide manner. **d**, FIDE and JSD scores of different methods for three domain counts (7, 10 and 15) on colon samples. **e**, Spatial domains assigned by Novae across two slides from the breast dataset (MERSCOPE and Xenium slides, see more details in 'Datasets used'). **f**, ARI comparison on the synthetic dataset with $n = 5$ seeds used for each model. Box plots indicate mean (center line), 25th and 75th percentiles (bounds of box), minima and maxima (whiskers), and all points shown individually. **g**, Same as **f** but for the FIDE score. **h**, Runtime comparison of the different methods (including training, batch-effect correction and domain assignment) across different dataset sizes (top). The difference is mostly due to the batch-effect correction and domain assignment (bottom), which is slow for Harmony/Leiden/mclust, on which Novae does not depend. **i**, UMAP of spatial representations, colored by slide, for each spatial-based method. Each dot represents the neighborhood representation of the cell, not the cell representation itself.

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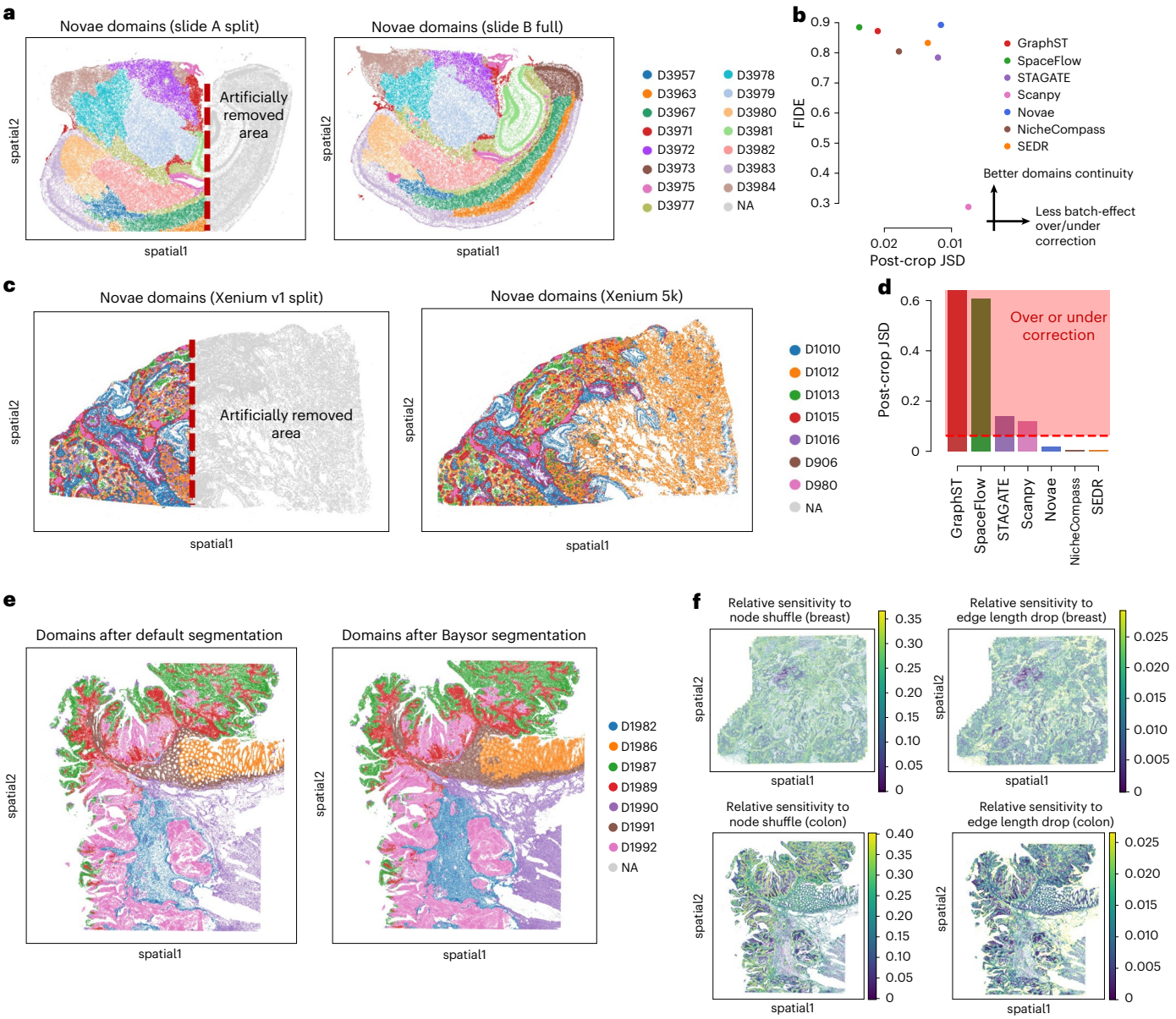


Fig. 4 | Novae robustness to perturbations: missing domains, segmentation change and input perturbation. a, Novae spatial domains for the mouse brain slides. Slide A (split) is on the left, and slide B (complete) is on the right. **b**, FIDE score of the different methods for the mouse brain slides and JSD score after cropping the second slide (compared to the first split slide). The models were trained on one split slide and one complete slide, but evaluated after splitting the second one. **c**, Novae spatial domains for the lung dataset (Xenium v1 split on the left, Xenium 5k on the right). **d**, JSD between the Xenium v1 and Xenium 5k slides after cropping the second slide. A high JSD, above the red line, means either under or over batch-effect correction. **e**, Comparison of the Novae spatial domains over two segmentation methods (the default 10x Genomics segmentation (left) and Baysor[34] (right)). **f**, Relative sensitivity of Novae to node shuffle and edge-length drop over the breast and colon slides.

requiring clustering to be run multiple times. Novae can perform this very rapidly, thus easing the analysis of different resolutions, while the other methods require running a time-consuming clustering again. Actually, the dependency on Harmony and Leiden becomes the main time bottleneck for large datasets, that is, even longer than the model training itself. To show that, we also conducted a timing analysis to run (1) training; (2) inference; (3) batch-effect correction; and (4) clustering for the different methods as the cell count increased. All models were trained on an A100 GPU with an early-stopping mechanism that triggered after ten epochs of no improvement, allowing us to fairly assess each method's convergence speed. We included both the trained and zero-shot versions of Novae in our comparisons, with timings summarized in Fig. 3h (top). Notably, despite being a larger model, the speed gain due to not depending on Harmony and Leiden makes Novae faster than the other methods. The speed gap is even larger when Novae is used in zero-shot mode. Also, regarding random-access-memory (RAM) usage, Novae supports lazy loading (instead of storing the full graph dataset in memory, each subgraph is created on the fly before running through the model). This prevents an important amount of RAM from being dedicated to loading the dataset. This allowed us to train Novae on a dataset composed of nearly 30 million cells using a GPU with 40 GB of RAM (see 'Implementation and training details'). In addition, Novae runs on local subgraphs instead of running on a full slide, which is new for two reasons. First, the subgraphs are smaller and therefore require less memory. Second, it is possible to use a larger number of layers in the neural network without aggregating information from further environments. For instance, having a graph neural network of 16 layers operating on the full slide would lead to mixed information between

two cells at a distance of 300 microns, which is usually not intended. Instead, using local subgraphs ensures only aggregating information from local microenvironments. Practically, due to the latter implementations, the maximum video-random-access-memory (VRAM) required by a GPU is the model size (128 MB for our current model on Hugging Face) and one mini-batch of graphs that is generated on the fly (between 2 MB and 20 MB depending on hyperparameters). This GPU VRAM usage is independent of the AnnData dataset, which stays on the CPU, and whose RAM usage is therefore independent of Novae.

Robustness to missing domains and perturbations

Novae's batch-effect correction uses optimal transport[25] (see 'Prototypes and swapped assignment task') to align spatial domains across slides; however, as different slides may contain spatial domains in varying proportions (or even entirely different domains) we relaxed the alignment objective by allowing a subset of prototypes to remain unused within each slide. This prevents the artificial mapping of all domains onto every slide (see 'Pan-tissue prototypes'). To evaluate the robustness of this approach, we created a benchmark dataset using two slides of 2.5-month mouse brain tissue (denoted as slide A and slide B in Fig. 4a). While these slides should exhibit similar spatial domain proportions, we artificially altered slide A by removing certain spatial domains. We then trained the different models on both slide A-split and slide B-complete. After training, we split slide B the same way and computed the JSD and FIDE scores to compare the domain proportions of the corresponding split sections. Our goal was to verify that the split and complete slides retained distinct domain proportions due to the introduced perturbation, while still expecting the proportions within the matching halves to align.

An overcorrection would result in artificially identical domain proportions across the slide A-split and slide B-complete, leading to major discrepancies after splitting slide B and comparing it to slide A-split. Conversely, under-correction (with slide-specific domains) would preserve distinct domain proportions, yielding a high JSD score. The results, summarized in Fig. 4b, show that Novae achieves both a high FIDE score (indicating domain continuity) and a low post-split JSD (indicating that Novae is not overcorrecting). The spatial domains of the other methods are shown in Supplementary Fig. 11. To further assess robustness, we created a second benchmark dataset using two adjacent Xenium slides (one processed with the Xenium v1 machine and the other with Xenium Prime 5k). Here, we altered the Xenium v1 slide by removing half of its data (Fig. 4c). The results, presented in Fig. 4d, confirm that removing a large part of the domain D1012 is not affecting too much the proportions on the Xenium v1 slide. In the opposite, some methods such as GraphST, SpaceFlow, STAGATE and Scanpy show a high JSD, due to under-correction. Beyond these benchmarks, we tested Novae's robustness against other perturbations, such as variations in segmentation methods. In Fig. 4e, we compare Novae's results using both the default 10x Genomics default segmentation (staining-based) and Baysor[34] (transcript-based), demonstrating that Novae consistently identifies the same spatial domains despite methodological differences in the segmentation. An interesting thing to note here is that the other models are also able to perform this task (Supplementary Fig. 13). Therefore, while a good segmentation is crucial for cell-level tasks (for example, cell-type annotation or ligand–receptor analysis), it shows a lower importance for the spatial domains definition. Last, Fig. 4f examines the sensitivity of Novae's output to node shuffling in input graphs and to an important reduction in edge length/weights (set to 0.01). As we could expect, we observe that node shuffling primarily affects domain interfaces, while edge-length reduction impacts mostly stromal and sparse regions. Additionally, we tested Novae's robustness to increasingly degraded tissue slides and with extreme batch effect. The experiments can be found in Supplementary Fig. 17 (showing performance drops at a slide degradation of 60%) and Supplementary Fig. 19. More details can be found in 'Perturbation analysis'.

Fluorescence spatial mapping of immune populations

To assess Novae's capabilities in analyzing fluorescence data, we applied it to a proprietary head and neck cancer dataset[35] generated with the CosMx[7] protein assays, which we made public for this paper. The methods describing Novae's usage on proteins are detailed in 'Novae usage on protein expression'. In this dataset, eight patients were stratified based on the presence of multinucleated giant cells (MGCs), which have been associated with favorable prognoses in both treatment-naive and preoperative chemotherapy-treated patients. Two of these slides are shown in Fig. 5a with their corresponding Novae spatial domains. While the previous study[35] had noted an increase in central memory $CD4^+$ T (T_{CM}) cells and plasma B cells in MGC-positive cases, the spatial organization of these populations remains unstudied. Novae revealed that these cell types, which showed an increase in patients with MGCs[35], predominantly localize within domain D492 (Fig. 5c), suggesting a potential coordinated immune response.

Additionally, Novae identified a significant reduction in domain D497 in patients who were MGC-positive (Fig. 5b), a region characterized by a stromal-like environment enriched in *FOXP3* T cells (regulatory T (T_{reg}) cells), indicating a potential decrease in immunosuppressive activity[36–38]. Although MGCs were observed accumulating around keratin bodies, no direct interactions with a specific immune cell type were detected. Furthermore, analysis of the overall macrophage population revealed a greater spatial separation from domain D495 in patients who are MGC-positive, which is enriched in $CD4^+$ and $CD8^+$ T cells (Fig. 5c). This suggests a potential diminishing immunosuppressive influence from tumor-associated macrophages in the presence of MGCs[39]. Also, regarding the MGC-mononuclear-phagocytes (MGC-MNP) cells, we observed a strong enrichment in domain D490, alongside MGC cells (Fig. 5c). This aligns with recent findings[35] reporting that MGC and MGC-MNP cells cluster within keratin-enriched niches, also known as granulomas. Second, domain D492 (enriched in B cells, plasma cells and macrophages) closely overlaps with lymphoid aggregates, which compose, among others, tertiary lymphoid structures (TLSs) identified on the hematoxylin and eosin (H&E)-stained image (Fig. 5a,c and Supplementary Fig. 20). This is consistent with known features of TLSs in head and neck squamous cell carcinoma, which are typically composed of these immune cell types[40]. Additionally, domain D437 is predominantly composed of adipocytes, which corresponds well to adipose-rich regions visible on the H&E stain (Supplementary Fig. 20). Finally, domain D497 shows a marked enrichment in endothelial and stromal cells (an expected association given that endothelial cells are largely localized within the stromal compartment of the tumor microenvironment[41]). Overall, these findings highlight Novae's ability to uncover spatial relationships between immune populations and tumor-associated features, providing deeper insights into the dynamic interplay of immune cells within the tumor microenvironment.

Incorporating histopathology information

In addition to spatial transcriptomics and/or proteomics, H&E-stained images can be integrated into Novae to provide complementary morphological context. Specifically, we compute patch embeddings over cell neighborhoods using a pathology foundation model such as CONCH[42]. These embeddings are then fused with Novae's graph-level representations to produce multimodal cell neighborhood embeddings (see 'Novae usage on multimodal spatial data with H&E'). We demonstrate this multimodal integration using a Xenium 5k human lung slide, for which a corresponding H&E image has been prealigned by 10x Genomics (see 'Datasets used'). In this sample, we compare three methodologies: (1) CONCH embeddings clustered with Leiden; (2) Novae domains derived from transcriptomics alone; and (3) multimodal Novae domains combining transcriptomic and H&E embedding. As shown in Fig. 5d, the multimodal Novae domains (right) delineate spatial structures more precisely than CONCH alone (left). Quantitatively, Fig. 5e shows that while Novae outperforms CONCH individually,

https://doi.org/10.1038/s41592-025-02899-6

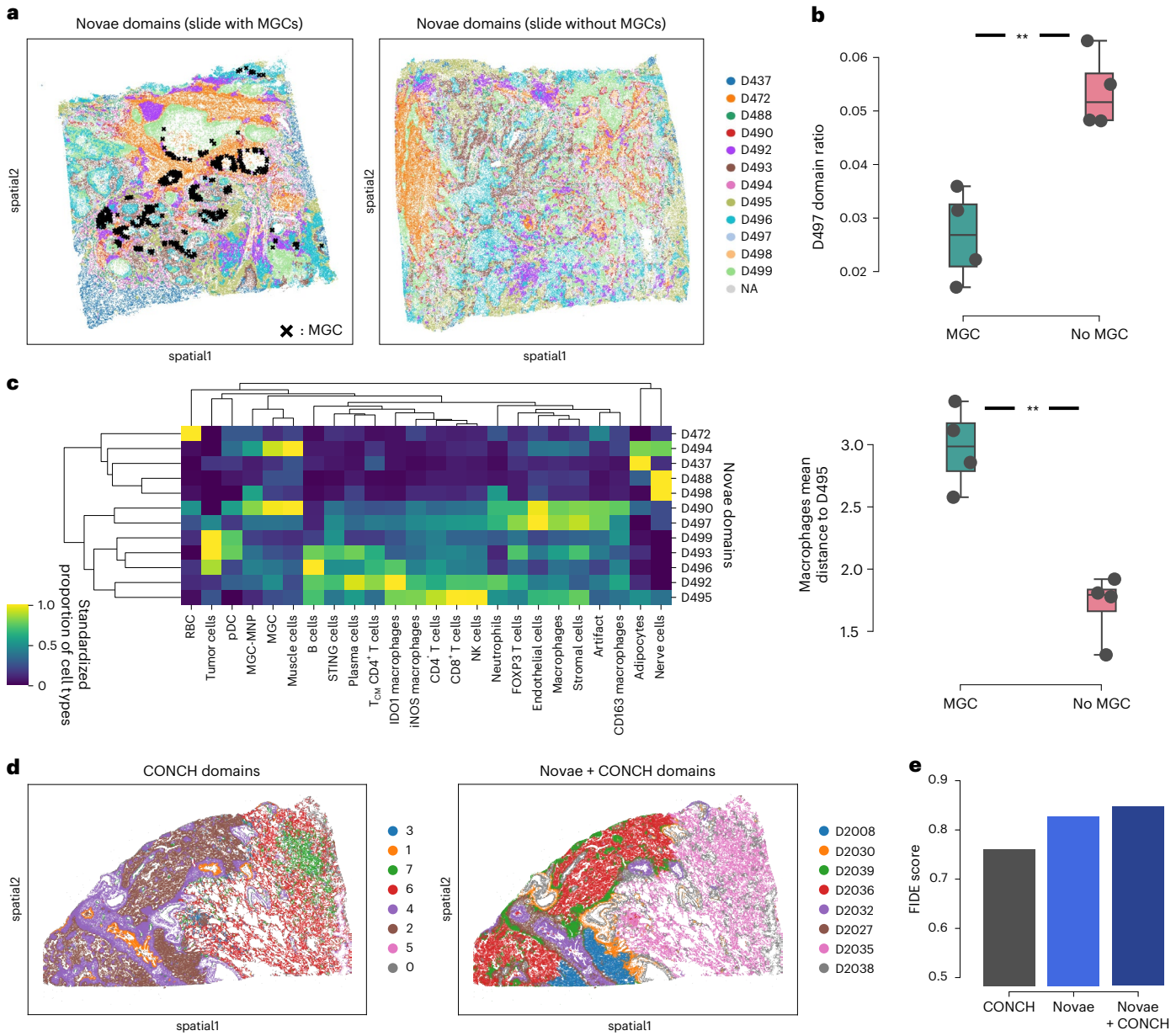


Fig. 5 | Novae spatial domains on other modalities with spatial proteomics (proprietary head and neck cancer dataset) and combined spatial transcriptomics and H&E (lung cancer). a, Spatial domain segmentation for two slides: patient with MGCs (left) and a patient without MGCs (right). **b**, Box plot comparing the proportions of cells in the domain D497 between $n = 4$ MGC-positive and $n = 4$ MGC-negative patients (two-tailed t-test P value of 0.0028), showing a reduction in MGC-positive cases (top). Box plot comparing macrophage distance to domain D495 (P value of 0.0010), indicating reduced interaction between macrophages and $CD4^+$/$CD8^+$ T cell-enriched regions in MGC-positive cases (bottom). Box plots indicate mean (center line), 25th and 75th percentiles (bounds of box), minima and maxima (whiskers) and all points are shown individually. $^{**}P < 0.01$. **c**, Heatmap of cell-type distributions across domains. The values are computed based on the proportions of cell types per spatial domain and then standardized across columns. RBC, red blood cells; pDC, plasmacytoid dendritic cells; NK, natural killer cells. **d**, CONCH domains (left) and multimodal Novae domains (right) on the human lung slide. **e**, FIDE score for CONCH, Novae, and Novae + CONCH on the human lung slide.

the fused Novae + CONCH model achieves the highest FIDE score, indicating improved performance through multimodal integration. Biologically, this added information enhances domain specificity. For instance, domain D2030, identified by multimodal Novae, is enriched in inflammatory monocytes (a distinction not captured by CONCH alone; Supplementary Fig. 16). Moreover, the latter heatmaps show that, while CONCH tends to underrepresent cell-type heterogeneity, Novae (with or without H&E) captures it more effectively. Also, the combination of Novae and CONCH enables the separation of domains D2032 and D2027, whereas Novae alone merges them into a single domain D2037 (Supplementary Fig. 16e). This improved resolution can be explained by the distinct morphology of region D2032, corresponding to the bronchus. Overall, adding the H&E information can add context and improve Novae's capabilities to identify heterogeneous spatial domains compared to the usage of spatial transcriptomics alone.

A multitude of downstream tasks

Novae's spatial representations enable the performance of multiple downstream tasks. To illustrate this, we applied three tasks across two datasets. The first dataset consists of a nondiseased lymph node and a reactive lymph node (shown in Fig. 6a–d), while the second dataset comprises six mouse brain slides collected at different time points, with

https://doi.org/10.1038/s41592-025-02899-6

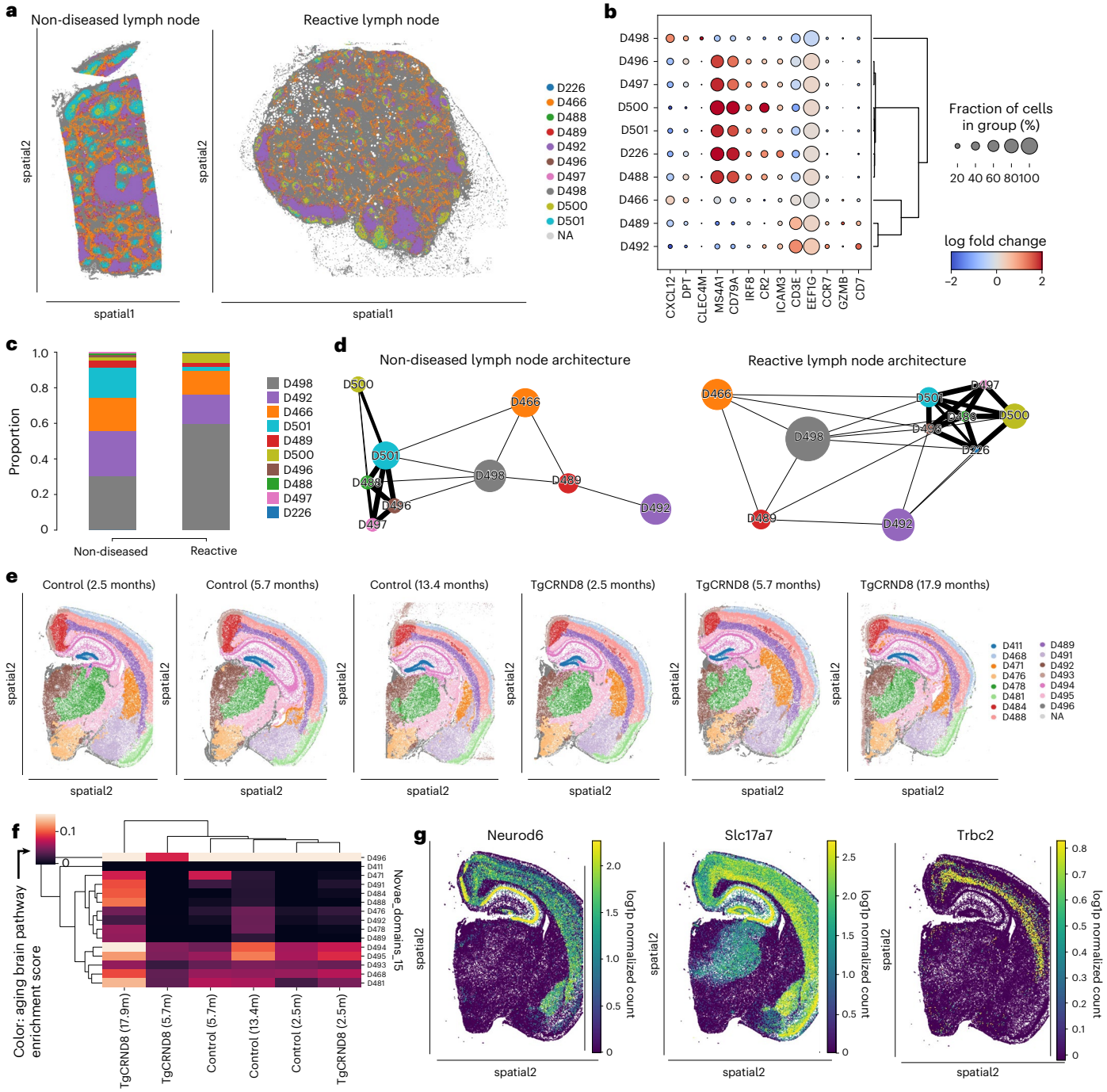


Fig. 6 | Downstream tasks examples on human lymph node and mouse brain slides. a, Spatial domains of the nondiseased (left) and reactive (right) lymph nodes. **b**, Heatmap of differentially expressed genes for the spatial domains of the reactive lymph node slide. **c**, Proportions of spatial domains for each lymph node slide. **d**, PAGA graphs comparing the domain organization of nondiseased (left) and reactive (right) lymph nodes. **e**, Spatial domains of mouse brain across different time points, comparing control conditions and Alzheimer-like pathology (*TgCRND8* samples). **f**, Heatmap of brain aging pathway activation across different domains and time points in the mouse brain slides. **g**, Expression patterns of three SVGs on the 2.5-month control mouse brain slide.

half exhibiting Alzheimer's-like pathology (further details in 'Datasets used'). For the lymph node dataset, spatial domains are displayed in Fig. 6a, with Fig. 6b,c providing details of gene expression and domain proportions to help the characterization of these spatial domains. To compare the spatial organization (or 'slide architecture'), users can apply trajectory inference methods to the spatial representations generated by Novae (more details in 'Downstream tasks examples'). As shown in Fig. 6d, we observed changes in the spatial organization of these domains. Notably, the D500 domain in the germinal center, enriched with ($CD79^+CR2^+$) mature B cells, is connected only to the D501 domain on the nondiseased slide, contrary to the reactive one, which is highly connected to five domains. Additionally, Fig. 6c reveals an inversion in the proportions of the D500 and D501 domains, also visible in Fig. 6a.

Regarding the mouse brain dataset, the spatial domains identified by Novae are presented in Fig. 6e. Although no substantial changes in brain architecture were observed (Supplementary Fig. 3 shows details), we do observe differences in brain aging pathway[43] enrichment that we computed for each spatial domain. As shown in Fig. 6f, the 17.9-month-old *TgCRND8* mouse, which exhibits Alzheimer's-like

pathology, shows higher brain aging, particularly in specific spatial domains such as D494 and D481. For example, the D494 domain has a high expression of the *Neurod6* gene, a key gene associated with brain aging[43]. This analysis highlights how identifying spatial domains enhances the understanding of pathway activation, demonstrating the utility of spatial domain identification in pathway analysis. In addition, differential gene expression analysis can be performed on cells grouped by spatial domains, allowing for the identification of SVGs. SVGs are genes whose expression varies the most across spatial domains. In Fig. 6g, we demonstrate this for the 2.5-month control mouse, showing the three most SVGs identified by Novae. Therefore, this helps better identify genes that have different activation levels depending on the spatial domain.

Discussion

The field of spatial transcriptomics is advancing rapidly, leading to the generation of increasingly larger datasets[44]. While this growth holds the promise of uncovering new insights that were difficult to achieve with scRNAseq[45], it also presents critical challenges in performing comparative spatial analyses across multiple slides. Current algorithms[8–11] are often designed for single or consecutive slide analyses and typically rely on external batch-effect correction methods, which may not be sufficient for the complexities of multislide spatial datasets. Novae addresses this gap by offering a more flexible solution that can operate across various tissues, conditions, technologies and gene panels. Compared to related models, we have shown that Novae can better identify cross-slide domains, while not overcorrecting the spatial domain assignments. Indeed, the flexibility introduced in the prototypes allows for certain domains to be tissue-specific or disease-specific. In addition, despite being a larger model, Novae achieves faster performance through two key advantages: (1) it operates independently of Harmony and Leiden, eliminating these computational bottlenecks, and (2) it offers a zero-shot mode that provides further speed improvements. This efficiency comes from Novae's native batch-effect correction, which is inherently spatially aware and eliminates reliance on external tools. Also, as a shared open-source model, Novae can be easily loaded and used for user-specific tasks without the need for hyperparameter optimization or large retraining.

Our study demonstrates Novae's potential, particularly when applied to large cohorts, to identify spatial domains enriched or specific to certain conditions or diseases. Using spatial trajectory analysis, we demonstrated how tissue organization and structure can be characterized, as shown by the comparison of a reactive lymph node with a nondiseased one. Lymph nodes are crucial for immune surveillance across different organs, with their structure enabling a targeted immune response during infections. In our analysis, we observed spatial reorganization in the reactive lymph node, notably the expansion of the D498 domain, which is enriched in the chemoattractant CXCL12, a key factor in recruiting inflammatory cells[46]. Concerning the second use case, we identified an enriched aging-related pathway in specific mouse brain regions linked to Alzheimer's-like pathology at defined time points, as well as brain-specific SVGs. Additionally, Novae effectively analyzed fluorescence data from a proprietary head and neck cancer dataset, revealing spatial relationships between immune cell populations in patients stratified by MGC presence. Notably, Novae identified key domain shifts, such as the colocalization of central memory $CD4^+$ T cells and plasma B cells in domain D492 and a significant decrease in the stromal-like, T_{reg} cell-enriched domain D497 in patients with MGCs, highlighting its potential to discover immunological dynamics related to clinical factors. Also, we have shown that we can incorporate H&E-derived embedding to enhance Novae's ability to identify and resolve complex, heterogeneous spatial domains. Overall, this study has showcased Novae's versatility in handling multiple downstream tasks, such as spatial domain identification, slide architecture and SVG analysis, all within a single framework.

Methodologically, Novae builds on the SwAV framework[24], a self-supervised learning technique originally designed for computer vision, which we adapt to graph-structured biological data. Unlike reconstruction-based methods such as variational autoencoders, SwAV does not require input reconstruction[24], making it better suited for clustering tasks and enabling the use of more expressive encoders. We introduce two key innovations: (1) adapting the SwAV objective to graph data with biologically relevant augmentations, and (2) operating on subgraphs, which allows us to train larger models without over-smoothing the expression over the whole slide. Notably, SwAV's use of optimal transport[25] for assignment (when one mini-batch contains one biological batch) enables native batch-effect correction, aligning spatial domains across samples without post hoc adjustments. We also use its prototype-based representation for efficient hierarchical clustering, as the learned prototypes serve as anchors for nested spatial domain definition. Together, these adaptations position SwAV as a powerful and computationally efficient framework for graph representation learning in spatial biology.

We also demonstrated that Novae can be trained on spatial proteomics data (for example, the CosMX Protein Assays[7], PhenoCycler[47] or MACSima[48]) or in a multimodal spatial context. Yet, the pretrained Novae models rely exclusively on spatial transcriptomics data. This limitation arises from the current absence of large-scale datasets combining spatial transcriptomics, spatial proteomics and histopathology on the same slides, which limits our capacity to train a foundation model across multiple spatial modalities. This lack of multimodal datasets also complicates the evaluation of multimodal models across diverse tissues and large cohorts of slides. In terms of robustness, we conducted a range of experiments to assess sensitivity to various perturbations, including segmentation changes, extreme batch effects and artificial slide degradation. While Novae exhibited strong stability against segmentation variation and batch effects, the degradation benchmark revealed a performance drop when approximately 60% of cells and gene expression were lost. Further investigations into sensitivity to segmentation changes could benefit from including tissues with extreme variations in cellular density. For example, densely packed tissues (where tightly clustered cells make segmentation more challenging) could be studied alongside sparser, lower-quality slides where cell boundaries are harder to define due to poor contrast or resolution. Additionally, because Novae operates on cell centroids, it is currently limited in its ability to represent complex tissue structures such as axons, which are not yet accurately segmented with current methods. Future segmentation advances that capture such structures could allow updating Novae's input graph to try solving this challenge. Looking ahead, incorporating a mixture-of-experts approach could enable joint analysis of both spot resolution and single-cell-resolution data within a unified framework. More broadly, expanding the size and heterogeneity of the training data would improve Novae's generalizability and its robustness to missing data domains (an area that remains a challenge). Addressing these limitations will be key to advancing Novae's utility and impact in spatial transcriptomics research.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41592-025-02899-6.

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Methods

Model input

In Novae, we consider a total of G genes, such as those in the human or mouse genome. A spatial transcriptomics slide captures the expression profile of a subset of these genes, denoted by $\mathcal{P} \subseteq \{1, \dots, G\}$, representing the indices of the genes in this specific panel. Throughout this article, we use the term 'cell' to refer to either a spot or a cell, although Novae is applicable to both spot resolution and single-cell resolution technologies. The expression data for a single slide is represented by a matrix $X = (\mathbf{x}_1, \dots, \mathbf{x}_N) \in \mathbb{R}^{N \times P}$, where N denotes the number of cells and $P = \mathrm{Card}(\mathcal{P})$ represents the number of genes in the panel $\mathcal{P}$. This matrix X contains normalized and logarithmized gene expression values. To incorporate spatial information, we utilize the two-dimensional localization of cells to construct a Delaunay graph for which edges longer than 80 microns are truncated. In this graph, nodes correspond to cells and edges indicate neighboring cells. The adjacency matrix $A \in \mathbb{R}^{N \times N}$ encodes the connectivity between cells, with edge weights representing the distances in microns between neighboring cells in the Delaunay graph. Also, an entire spatial transcriptomics dataset consists of multiple slides, each described by a tuple $(X, \mathcal{P}, A)$, where the size of each gene panel $\mathrm{Card}(\mathcal{P})$ can vary across different slides. Additionally, for a given slide and a specific cell i, we define a subgraph $\mathcal{G}_i$ consisting of cells within a distance of n_{local} edges from cell i; typically, we choose $n_{\text{local}} \in \{2, 3\}$, creating graphs of roughly 15–80 cells (see experiments in Supplementary Fig. 18). The features of a node (or cell) in this subgraph are the gene expression vectors of the corresponding cells. For instance, the feature vector for cell i is $\mathbf{x}_i \in \mathbb{R}^P$. In the following sections, $\mathcal{G}_i$ is used to obtain a spatial domain representation for the subgraph origin cell i.

Augmentation

Augmentation is a technique used to introduce artificial variations into the data, enhancing the dataset's diversity. This approach is commonly employed to improve the robustness and generalization abilities of models. In the context of spatial transcriptomics, we apply two biologically relevant augmentations to strengthen the robustness of Novae.

First, we introduce a 'pseudo batch effect' noise to reduce the model's sensitivity to batch effects. For a given subgraph input, we sample two vectors: $\mathbf{a} \sim \text{exponential}(\lambda)^P$ and $\mathbf{s} \sim \text{Normal}(0, \sigma^2 I_P)$, where λ and σ are hyperparameters, and I_P is the identity matrix of size $P \times P$. The gene expression vector for each cell i is then updated as $\mathbf{x}_i^{(\text{noise})} = \mathbf{a} + \mathbf{x}_i \odot (1 + \mathbf{s})$.

Second, we randomly subset the gene panel according to a ratio $\gamma \in [0, 1]$. For a given panel $\mathcal{P}$, we select $\lfloor \gamma P \rfloor$ genes, resulting in a new panel $\mathcal{P}' \subset \mathcal{P}$. This augmentation simulates the effect of different panels of genes, as if multiple machines generated the data, or as if the panel was updated during a study. After applying these augmentations, each cell is represented by the expression vector $\mathbf{x}_i^{(\text{augmented})} = (x_{ij}^{(\text{noise})})_{j \in \mathcal{P}'}$. This augmented representation helps the model better generalize across varying conditions and datasets.

Cell embedding

After augmentation, cells are transformed into panel-invariant embeddings, which serve as the node features for the graph encoder described in the subsequent section. Let $\mathbf{v}_1, \dots, \mathbf{v}_G \in \mathbb{R}^E$ denote a list of trainable gene embeddings, where E represents the embedding size. For a given gene panel $\mathcal{P} \subseteq \{1, \dots, G\}$, we consider only the embeddings corresponding to the genes in this panel. These embeddings are L2-normalized and then multiplied by the cell's scaled gene expression vector $\mathbf{x} \in \mathbb{R}^P$. Specifically, the embedding for a cell i is calculated as follows:

$$\mathrm{embed}(\mathbf{x}_i, \mathcal{P}) = \frac{\sum_{j=1}^{P} x_{ij} \mathbf{v}_{\mathcal{P}[\mathbf{j}]}}{\sqrt{\sum_{j=1}^{P} \mathbf{v}_{\mathcal{P}[\mathbf{j}]}^2}} \in \mathbb{R}^E,$$

where the square root, square and division operations are performed element-wise. The purpose of the L2 normalization is to ensure that the embedding weights are comparable across different gene panel sizes. This can be compared to a principal-component analysis (PCA) reduction, where the components are trainable gene embeddings, and the L2 normalization ensures that each panel has consistent weights across different 'gene programs'. Specifically, for each $e \leq E$, $\mathbf{v}_{ge}$ represents the weight of gene $\mathbf{v}_g$ in the gene program e. Additionally, rather than training all gene embeddings from scratch, we can initialize with pretrained gene embeddings from scGPT[49]. To prevent domain shift for genes not present in the Novae dataset, these pretrained embeddings are frozen, and we introduce a trainable linear layer afterward. Thus, the updated cell embedding formula becomes

$$\mathrm{embed}(\mathbf{x}_i, \mathcal{P}) = \frac{\sum_{j=1}^{P} x_{ij} (\mathbf{W}\mathbf{v}_{\mathcal{P}[\mathbf{j}]} + \mathbf{b})}{\sqrt{\sum_{j=1}^{P} (\mathbf{W}\mathbf{v}_{\mathcal{P}[\mathbf{j}]} + \mathbf{b})^2}} \in \mathbb{R}^E,$$

where $\mathbf{W} \in \mathbb{R}^{E \times E}$ and $\mathbf{b} \in \mathbb{R}^E$ are trainable parameters. During training, the function $\mathrm{embed}(\cdot, \cdot)$ is applied to the augmented gene expression vector $\mathbf{x}_i^{(\text{augmented})}$, whereas during prediction or inference, it is applied to the original gene expression vector $\mathbf{x}_i$.

Graph encoder

The graph encoder utilized in Novae is a Graph Attention Network[16] (GAT), which employs attention mechanisms to aggregate information from neighboring cells. The GAT is composed of multiple layers, with each layer potentially having multiple attention heads. The input to the GAT is a subgraph $\mathcal{G}$, where node features are embedded cell features (as described in 'Cell embedding'). For each cell i, the initial node feature is $\mathbf{h}_i^{(0)} = \mathrm{embed}(\mathbf{x}_i^{(\text{augmented})}, \mathcal{P}')$ during training, and $\mathbf{h}_i^{(0)} = \mathrm{embed}(\mathbf{x}_i, \mathcal{P})$ during inference. For each layer l, the node features for the next layer are calculated as:

$$\mathbf{h}_i^{(l+1)} = \sum_{j \in \mathcal{N}(i) \cup \{i\}} \alpha_{i,j}^{(l)} \mathbf{\Theta}_t^{(l)} \mathbf{h}_j^{(l)},$$

where the attention coefficients $\alpha_{i,j}^{(l)}$ are defined as:

$$\alpha_{i,j}^{(l)} = \frac{\exp\left(\mathbf{a}^{(l)\top} \mathrm{LeakyReLU}\left(\mathbf{\Theta}_s^{(l)} \mathbf{h}_i^{(l)} + \mathbf{\Theta}_t^{(l)} \mathbf{h}_j^{(l)} + \mathbf{\Theta}_e^{(l)} A_{ij}\right)\right)}{\sum_{k \in \mathcal{N}(i) \cup \{i\}} \exp\left(\mathbf{a}^{(l)\top} \mathrm{LeakyReLU}\left(\mathbf{\Theta}_s^{(l)} \mathbf{h}_i^{(l)} + \mathbf{\Theta}_t^{(l)} \mathbf{h}_k^{(l)} + \mathbf{\Theta}_e^{(l)} A_{ik}\right)\right)}.$$

In these equations, $(\mathbf{a}^{(l)}, \mathbf{\Theta}_s^{(l)}, \mathbf{\Theta}_t^{(l)}, \mathbf{\Theta}_e^{(l)})_{1 \leq l \leq L}$ denotes parameters of the model (biases or matrices) and $\mathcal{N}$ represents the set of neighbors of a cell in the subgraph $\mathcal{G}$. After L sequential layers, each node i has a feature vector $\mathbf{h}_i^{(L)}$. Then, to obtain a unified representation for the entire subgraph $\mathcal{G}$, we employ an attention aggregation layer, resulting in a graph-level representation $\mathbf{z} := \text{attention} - \text{aggregation}((\mathbf{h}_i^{(L)})_i)$. The subgraph representation $\mathbf{z}$ is learned through the self-supervised task described in 'Prototypes and swapped assignment task'. Note that, for simplification, the above equations detail the graph encoder with only one head. With multiple attention heads, the GAT computes separate attention coefficients and node features for each head. The outputs from all heads are then concatenated (or averaged) to form the final node representation for each layer, allowing the model to capture diverse aspects of the node's neighborhood.

Prototypes and swapped assignment task

As the dataset lacks ground truth, we need to train Novae using an unsupervised approach. Specifically, as we aim to pretrain a foundation model, we will leverage self-supervised learning[15,50], which is well-suited for capturing meaningful data representations. Among the different self-supervision frameworks, SwAV[24] is a self-supervised learning algorithm that integrates contrastive learning and clustering.

The main concept is to learn representations by predicting cluster assignments derived from different augmentations (views) of the same image. In our context, instead of using two views of the same image, we utilize two closely related subgraphs of cells. Specifically, we select a pair (i, j) of cells separated by n_{view} edges, from which we derive the corresponding subgraphs $(\mathcal{G}_i, \mathcal{G}_j)$. When n_{view} is small enough (typically 2 or 3; Supplementary Fig. 18), we expect that the subgraph representations $(\mathbf{z}_i, \mathbf{z}_j)$ will belong to the same spatial domain. As these subgraphs are based on a cell of origin, we use interchangeably the terms (1) subgraph representation $\mathbf{z}_i$ of the subgraph $\mathcal{G}_i$ and (2) the neighborhood representation of the cell i itself. To actually assign these representations to spatial domains, we train a set of K vectors $\mathbf{c}_1, \ldots, \mathbf{c}_K \in \mathbb{R}^O$ in the unit sphere ($\forall k, \|\mathbf{c}_k\|_{L^2} = 1$, where $\|\cdot\|_{L^2}$ is the L2-norm). These trainable embeddings are referred to as prototypes in the original SwAV[24] paper. In our application, they represent a high-resolution spatial domain (in practice, K is large). The representation $\mathbf{z}_i$ of each subgraph is projected onto the prototypes, over which a softmax function is applied:

$$\mathbf{p}_i = \left(\frac{\exp\left(\frac{1}{\tau} \frac{\mathbf{z}_i^T}{\|\mathbf{z}_i\|_{L^2}} \mathbf{c}_k\right)}{\sum_{1 \le k' \le K} \exp\left(\frac{1}{\tau} \frac{\mathbf{z}_i^T}{\|\mathbf{z}_i\|_{L^2}} \mathbf{c}_{k'}\right)} \right)_{1 \le k \le K} \in \mathbb{R}_+^K,$$

where τ is a temperature parameter that controls the sharpness of the softmax distribution. Intuitively, p_{ik} represents the probability that the neighborhood representation of cell i belongs to the prototype k. Again, to avoid confusion, we remind that the neighborhood representation of a cell i is assimilated to the representation $\mathbf{z}_i$ of its corresponding subgraph $\mathcal{G}_i$. To prevent the representations from collapsing into a single prototype, we define a 'corrected' assignment $\mathbf{q}_i$ that aligns with the distribution $\mathbf{p}_i$ while considering global mini-batch statistics. Essentially, we aim for each mini-batch to represent all prototypes as evenly as possible. This assignment $\mathbf{q}_i$ is derived from the result of an optimal transport[25] (OT) problem over a mini-batch of size B (with each mini-batch being dedicated to one slide). Specifically, given a mini-batch of B subgraph representations $Z = (\mathbf{z}_1, \ldots, \mathbf{z}_B) \in \mathbb{R}^{B \times O}$ and the matrix of all prototypes $C = (\mathbf{c}_1, \ldots, \mathbf{c}_K) \in \mathbb{R}^{K \times O}$, the OT problem is defined as:

$$Q^* := \underset{Q \in \mathcal{Q}}{\operatorname{argmin}} \left(\operatorname{Tr}(QCZ^\top) - \epsilon H(Q)\right),$$

where H is the Shannon entropy, ϵ is a regularization hyperparameter, and $\mathcal{Q}$ is the transportation polytope defined by $\mathcal{Q} = \left\{Q \in \mathbb{R}_+^{B \times K} | Q\mathbf{1}_K = \frac{1}{B}\mathbf{1}_B,\ Q^\top \mathbf{1}_B = \frac{1}{K}\mathbf{1}_K\right\}$. This OT problem ensures a smooth distribution of assignments across the different prototypes, preventing mode collapse. It also helps avoid learning 'artificial' prototypes that are specific to certain slides or batch effects. The Sinkhorn–Knopp algorithm is employed to approximate Q^*, as detailed in Supplementary Notes. Subsequently, we obtain $\mathbf{q}_i = Q_i^* \in \mathbb{R}^K$, which represents the 'corrected' assignments for cell i. The loss function used in Novae is a cross-entropy applied to the 'swapped' prediction problem. Specifically, $\mathbf{q}_i$ serves as the 'self-supervised ground truth' for $\mathbf{p}_j$, while $\mathbf{q}_j$ is used for $\mathbf{p}_i$. We formally define the loss as:

$$\mathcal{L}(\boldsymbol{\theta}, i, j) = -\sum_{k=1}^{K} \left(q_{ik} \log p_{jk} + q_{jk} \log p_{ik}\right),$$

where $\boldsymbol{\theta}$ represents all the model parameters and (i, j) denotes a pair of cells separated by n_{view} edges. This loss function encourages the soft spatial domains of cell i to resemble the soft 'pseudo-ground truth' spatial domains of cell j, and vice versa. The model is trained via back propagation of this loss using the Adam optimizer. Note that the gradients are detached during the optimal transport[25] part, meaning that the loss is backpropagating via $(\mathbf{p}_i, \mathbf{p}_j)$.

Pan-tissue prototypes

The loss defined in the previous section is limited when training on multiple tissues, as the optimal transport problem enforces similar distribution assignments to the prototypes across all slides. However, many tissues possess unique, nonoverlapping spatial domains, making it biologically unrealistic to train shared prototypes for all tissues. To address this, we introduce a slide-by-prototype matrix of weights, allowing the model to not map a slide to all domains. Specifically, we initialize a queue $W^{(\text{queue})} \in \mathbb{R}^{S \times \text{size} \times K}$, where S is the number of different slides, size is the queue size and K is the number of prototypes (filled with 1 everywhere). For each slide and mini-batch of size B (the mini-batch contains cells of the same slide), we calculate the maximum assignment probability for each prototype, $(\max_{i=1}^{B} p_{ik})_{1 \le k \le K} \in \mathbb{R}^K$, and store this in the queue at the corresponding slide's position. Technically, this involves rolling the queue weights for a specific slide s and updating the entry as $W_{0s}^{(\text{queue})} \leftarrow (\max_{i=1}^{B} p_{ik})_{1 \le k \le K}$. Afterward, we compute the maximum probabilities over the queue size, that is the second dimension, which we denote $W^* \in \mathbb{R}^{S \times K}$. This slide-by-prototype matrix W^* is used for each mini-batch to subset the prototypes on which the optimal transport is performed to define the corrected assignments $\mathbf{q}$. For a given slide $s \in \{1, \ldots, S\}$, the codes are computed using the prototypes k for which $W_{sk}^* > 0.9 \max_{j=1}^{S}(W_{jk}^*)$, meaning that the probability of having this prototype is high enough (compared to the slide which most likely have this domain). Yet, to not only learn slide-specific prototypes, we ensure that each slide is mapped over at least ρK prototypes, where $\rho \in [0, 1]$ is a hyperparameter. If less than ρK prototypes match the above condition, we choose the $\lfloor \rho K \rfloor$ prototypes with the highest W_{sk}^* score. Typically, we choose $\rho \gg \frac{1}{S}$ to ensure that prototypes will be shared by multiple slides. Indeed, we expect slides from the same tissue to have shared prototypes (or 'tissue-specific' prototypes), while some prototypes should be shared across multiple tissues. The covariance of the prototypes for the human run is shown in Supplementary Fig. 1, and we show UMAP representations of prototypes in Supplementary Fig. 6.

Assignment to spatial domains

Prototypes can be considered as centroids of elementary spatial domains. As the desired number of spatial domains may vary, we employ hierarchical clustering on the prototypes. In this setup, the prototypes serve as the leaves of a hierarchical tree, with each level of the tree representing increasingly coarse-grained spatial domains. For a given subgraph representation $\mathbf{z}_i$, we assign it to the closest prototype by selecting the one with the highest dot product, defined as $C_i^{(\text{leaf})} := \arg\max_{k=1}^{K} \mathbf{z}_i^T \mathbf{c}_k$. This assignment $C_i^{(\text{leaf})}$ indicates to which tree-leaf the cell i is associated. At a particular level l of the tree, the spatial domain assignment for cell i is determined by $C_i^{(l)} = \text{Map}(l, C_i^{(\text{leaf})})$, where $\text{Map}(l, \cdot)$ is a mapping function that associates the leaf prototype $C_i^{(\text{leaf})}$ with a cluster at level l (the level l is chosen according to the number of desired clusters). This hierarchical approach allows each cell representation to be assigned to spatial domains of varying resolutions efficiently (constant time). In other words, the prototypes serve as elementary spatial domains, used to define spatial domains at the desired resolution. In addition to the hierarchical-based assignment, we offer the possibility to use Leiden[13] on the prototypes. While this does not keep the hierarchy property, it has the benefit of depending on a resolution parameter instead of a number of spatial domains. Therefore, this option can sometimes be preferred to the hierarchical clustering of the prototypes, depending on the user use cases. This prototype Leiden clustering is also very fast, as the number of prototypes (typically 512) is very small compared to the number of cells. Finally, the cell assignment to the spatial domain also uses an arg max and a mapping, as described above for the hierarchical clustering.

Zero-shot and fine-tuning

We train Novae on a large dataset composed of 18 different tissues, as detailed in 'Datasets used'. Therefore, we can save this model on a hub

(in particular, the Hugging Face Hub), and then anyone can download this model and reuse it on their own dataset without retraining. For that, for a given new dataset, we compute the spatial domain representation $\mathbf{z}$ of all cells in all slides. Afterward, we run a K-means algorithm from sklearn[51] on the representations to define K centroids, which will act as the new prototypes. These prototypes are used for spatial domain assignment as detailed in the previous section. We denote this as zero-shot because Novae is not retrained. For fine-tuning, we apply the same approach but then retrain the model for a few epochs, potentially one only epoch.

Batch-effect correction

The optimal transport approach ensures the retrieval of relatively similar spatial domains across slides of the same tissue. Therefore, the spatial domain assignments C_i^l are corrected for the different slides, meaning that correcting the actual representations $\mathbf{z}$ is not always necessary. If correcting the representation is actually desired, for each spatial domain we compute the centroid representation of the slide that has the most cells in this spatial domain, which generates a list of centroids $\mathbf{z}_0^{(\text{centroid})}, \dots, \mathbf{z}_L^{(\text{centroid})} \in \mathbb{R}^O$. Then, for each slide and for each spatial domain l, we compute the mean representation $\bar{\mathbf{z}}_l$, and the representation of a cell i in spatial domain l becomes $\mathbf{z}_i^{(\text{corrected})} = \mathbf{z}_i - \bar{\mathbf{z}}_l + \mathbf{z}_l^{(\text{centroid})}$. Essentially, we translate the representations to align the centroids to the reference centroid.

Novae usage on protein expression

By default, Novae was trained on a large spatial transcriptomic dataset, but we can also use it on spatial proteomics data such as PhenoCycler[47] or MACSima[48]. Instead of using the scGPT gene embeddings, we use the components of a PCA fitted on the cell-by-protein expression table for the dataset of interest. The latter table can be extracted by averaging the mean intensity of all proteins over all the cell masks, for instance via Sopa[23]. Afterward, we can train Novae as for spatial transcriptomic dataset, with the protein embeddings behaving similarly to the gene embeddings.

Novae usage on multimodal spatial data with H&E

We extended Novae to integrate spatial transcriptomics with the H&E imaging modality. To achieve this, we first partition the H&E slide into overlapping patches and compute visual embeddings for each patch using a foundation model such as CONCH[42]. For computational efficiency, we retain only patches that contain at least one cell. The resulting embeddings are then reduced to their top 50 principal components using scikit-learn[51], followed by standardization. Each cell is subsequently assigned the reduced embedding of its nearest H&E patch. This preprocessing occurs before training Novae. Technically, to align modalities in space, we adopt SpatialData[22] as our data structure, and apply Sopa[23] to compute these patch-level embeddings. In Novae, we added a small wrapper to Sopa to map each cell to the closest patch and save its PCA representation of the patch embeddings. Then, during training, for each subgraph, we retrieve the H&E embedding associated with the origin cell and concatenate it with the output of the graph attention encoder. The combined representation is then passed through a three-layer perceptron. The rest of the methodology remains unchanged. Finally, this multimodal usage can be similarly applied to spatial proteomics combined with H&E, substituting transcript expression for protein expression. In cases where all three modalities (spatial transcriptomics, spatial proteomics and H&E) are available, we recommend concatenating the transcriptomic and proteomic profiles into a unified 'spatial' modality at the single-cell level before integrating the H&E features as described above.

Downstream tasks examples

For downstream tasks, we either use the spatial representations of Novae (vectors), or directly, the spatial domain assignments per cell (categories). For instance, the SVGs are defined by the `scanpy.tl.rank_genes_groups` function from Scanpy[30] over the categorical spatial domains of Novae, and increasing the resolution of the Novae domains results in more local SVGs. Regarding pathways, we use the `scanpy.tl.score_genes` function from Scanpy applied to gene sets from the Gene Set Enrichment Analysis database[52,53]. This gives a score for every cell and every pathway. Afterward, we average these scores per spatial domain, resulting in a domain-by-pathway heatmap. We plot this heatmap using the `seaborn.clustermap` function from Seaborn[54], which groups domains with similar patterns of scores across pathways. More specifically, in Fig. 6, we used the LEE_AGING_CEREBELLUM_UP[43] pathway (https://www.gsea-msigdb.org/gsea/msigdb/mouse/geneset/LEE_AGING_CEREBELLUM_UP.html?ex=1). Regarding trajectory inference, we can run any method on the representations of Novae, grouped by categorical spatial domain. In practice, in this article, we used PAGA[32] and more specifically the `scanpy.tl.paga` implementation from Scanpy.

Metrics for model comparison

The FIDE score is used to quantify the spatial domain continuity. More specifically, let $C = (C_i)_{1 \le i \le N}$ be the categorical spatial domain predictions for N cells of a slide. The FIDE score is defined as $\text{FIDE}(C, A) = \text{F1} - \text{score}((C_i, C_j)_{i,j \text{ s.t. } A_{ij} > 0})$, where A_{ij} is positive when the cells i and j are graph neighbors. Intuitively, for an edge $i \leftrightarrow j$, the edge is an inter-domain edge if $C_i \neq C_j$. A high number of inter-domain edges shows a low domain continuity, therefore quantified by a low FIDE score. On multiple slides, we average the per-slide FIDE scores. The JSD is used to quantify the homogeneity of domains across multiple slides. For each slide $s \in [[1, S]]$, we compute $\boldsymbol{\pi}_s \in [0, 1]^D$, the proportion of each domain among the slide, where D is the total number of spatial domains. Then, the JSD metric is defined as $\text{JSD}(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_S) = H(\frac{1}{S}\sum_{s=1}^{S} \boldsymbol{\pi}_s) - \frac{1}{N}\sum_{s=1}^{S} H(\boldsymbol{\pi}_s)$, where H is the Shannon entropy. The ARI[33] is defined as the percentage of concordance between two clusterings. Mathematically, $\text{ARI} = \frac{\text{TP+TN}}{\text{TP+FP+FN+TN}}$, where TP is the number of true positives, TN the number of true negatives, FP the number of false positives and FN the number of false negatives.

Datasets used

In total, 78 public spatial transcriptomics slides were collected from single-cell resolution technologies (Xenium, MERSCOPE and CosMX). The datasets are available on the vendor websites, that is, respectively 10x Genomics, Vizgen and Nanostring. This dataset covers 18 tissues (human bone marrow, human brain, human breast, human colon, human kidney, human liver, human lung, human lymph node, human ovarian, human pancreas, human prostate, human skin, human tonsil, human uterine, mouse femur, mouse brain, mouse colon and whole mouse) and a total of 28,909,516 cells. It consists of 853,622 cells from the CosMX technology, 10,196,835 cells from the MERSCOPE technology and 17,859,059 cells from the Xenium technology. The CosMX covers 2 different tissues, 8 for the MERSCOPE and 17 for the Xenium technology. In total, eight tissues contain slides from multiple technologies (breast, colon, liver, lung, ovarian, pancreas, prostate and skin). For the benchmark, we selected five colon slides from the Xenium technology, divided into three distinct gene panels. The first panel includes 325 genes and comprises 270,984 cells, the second panel contains 422 genes and 924,597 cells and the third panel consists of 480 genes, with a total of 388,175 cells. The slides can be found in the Novae dataset (Data Availability) and their names are listed in Supplementary Data 1. We also selected two breast slides from different platforms: Xenium and MERSCOPE, containing a total of 1,290,084 cells. Again, their names are listed in Supplementary Data 1. Similarly, we listed in Supplementary Data 1 the name of the lymph node and mouse brain slides used in Fig. 6. Regarding the mouse brain slides, note that the 'Alzheimer-like pathology' concerns the CRND8 APP-overexpressing (*TgCRND8*) transgenic mouses, as detailed in https://www.10xgenomics.com/datasets/

xenium-in-situ-analysis-of-alzheimers-disease-mouse-model-brain-coronal-sections-from-one-hemisphere-over-a-time-course-1-standard. The multimodal example was performed on the following 10x Genomics slide (https://www.10xgenomics.com/datasets/xenium-human-lung-cancer-post-xenium-technote). For the MGC analysis, we used the CosMX protein dataset from a previous study[35] consisting of eight head and neck samples (see the original article for a full description[35]). Slides were selected so that half contained MGCs, and the dataset was processed using the Sopa pipeline[23]. Additionally, we created a synthetic dataset to compare the different models to a ground truth. More specifically, seven domains were generated in a circular manner, for five different slides with 100 genes. For each domain of each slide, the gene expression was sampled from an exponential law whose parameter is the sum of a slide-specific parameter and a domain-specific marker. For more details, see Supplementary Notes.

Spatial domains assignment benchmark

In our benchmark study, we evaluated the performance of our method, Novae, against six state-of-the-art methods: SpaceFlow[10], GraphST[9], STAGATE[8], SEDR[11], NicheCompass[29] and Scanpy[30] (which uses a PCA followed by Leiden clustering), using three spatial transcriptomics datasets: the colon dataset, the breast dataset and a synthetic dataset, as described in 'Datasets used'. Regarding the colon dataset, each method was applied independently to the different gene panels (except Novae, which works across different gene panels). For the breast dataset, we focused on the intersection of the two slides, which included 185 common genes, ensuring that all methods were applied consistently to this shared gene set. Again, this excludes Novae, which can work on both gene panels at the same time. After training the different models on each dataset, we applied Harmony[12] to address potential batch effects introduced by the use of different slides or experimental conditions, aligning the spatial transcriptomic data across slides. Following Harmony, we used the mclust[14] algorithm for clustering the representation, each cluster being considered as a spatial domain. Note that clustering was performed on the concatenated data to get consistent clusters across slides. Again, the usage of Harmony and mclust does not concern Novae, which includes both operations natively, as described in 'Batch-effect correction' and 'Assignment to spatial domains'.

Perturbation analysis

To assess the robustness of Novae to different perturbations, as shown in 'Robustness to missing domains and perturbations', we created multiple datasets. First, for the missing domains analysis, we took two mouse brain slides (Supplementary Data 1). The first slide was split based on the spatial coordinates $x < 4{,}000$, and the second one was split based on a polygon selection to match as much as possible the split of the first (Supplementary Fig. 12). Regarding the second benchmark, we took consecutive slides of lung data (https://www.10xgenomics.com/datasets/xenium-human-lung-cancer-post-xenium-technote), one with the Xenium v1 and one with the Xenium Prime 5k panel. The Xenium v1 slide was split based on coordinates $x < 5{,}000$. Based on the predicted Novae domains, we established a threshold for detecting overcorrection. To define this threshold, we considered that, in a case of maximal overcorrection, the Xenium v1 split sample would match the proportions of the full Xenium 5k sample. Therefore, to simulate a scenario of maximal overcorrection, we compared the full Xenium 5k sample with its split version of the exact same slide. Indeed, in this scenario of maximal overcorrection, we expect the Xenium 5k sample (complete) to have the same domain proportions as the Xenium v1 split, so compare the Xenium 5k (complete) to the Xenium 5k (split) mimics a prediction of perfect overcorrection. The computation of the JSD score of this approach gives us a threshold for identifying excessive correction. For these benchmarks, we used $\gamma = 0.3$, which is the default value for a new Novae model. Second, for robustness to segmentation methods, we used a colon sample (Supplementary Data 1), which was originally segmented with Cellpose, and we ran Baysor[34] using Sopa[23] to have another segmentation over the same slide. To compare the overlap between the spatial domains obtained from both slides, we expanded the cell's centroid by a radius of 10 microns, transforming each spatial domain into a multipolygon. For each spatial domain, we computed the intersection-over-union of the multipolygons obtained for both segmentation methods (Supplementary Fig. 13). Third, we computed the relative sensitivity of Novae's output based on $\frac{||\mathbf{y}_1 - \mathbf{y}_0||_2}{||\mathbf{y}_0||_2}$, where $\mathbf{y}_1$ is the Novae output post-perturbation for one subgraph, and $\mathbf{y}_0$ is the Novae output before the perturbation for the same subgraph. We studied two perturbations, first a node shuffle, and second, setting the edge length to 0.01. Finally, we also averaged the attention of Novae across the layers, then applied a softmax with temperature 0.01, and computed the entropy of the aggregated attention (Supplementary Fig. 10). Also, to evaluate Novae's robustness to batch effects and consistency of spatial domain assignments, we implemented a systematic perturbation approach using inter-slide batch effect simulation. We selected representative spatial transcriptomics datasets from lung and brain tissues as test cases, generating five progressively perturbed versions of each slide to mimic realistic experimental batch effects encountered in multislide studies. Each perturbation level applied whole-slide modifications, including library size scaling with log-normal distribution (σ ranging from 0.02 for minimal to 0.5 for extreme effects), gene-specific efficiency changes affecting 3–50% of genes with varying capture rates, ambient RNA contamination targeting 5–100 randomly selected genes and spatial field gradients for stronger perturbation levels. Finally, to evaluate robustness to slide quality, we simulate degradation on a slide by randomly removing a proportion $\delta \in [0, 1]$ of cells and masking the expression of a proportion δ of genes. We then train Novae using the degraded slide alongside the corresponding reference (nondegraded) slide. After training, we compare the cell assignments in the degraded slide to the corresponding cells in the reference. This comparison yields an accuracy score that reflects the model's robustness to data degradation (Supplementary Fig. 17).

Implementation and training details

The input of Novae is composed of one or many AnnData[21] objects from the anndata[21] data structure of the scverse[20] community. They contain cell-by-gene tables, which can be typically obtained via (1) reading the proprietary data, for example, using the readers from SpatialData[22] or (2) performing a new segmentation, for example, using CellPose[55] or Baysor[34]. When constructing the Delaunay graph, edges longer than 100 microns were dropped, which ensures long-distance cells are not considered neighbors. We implemented Novae using Python and the deep learning framework Pytorch[56] as well as Pytorch Geometric[57]. Each mini-batch contains B subgraphs such that all these subgraphs come from the same slide. For RAM efficiency, the subgraphs are lazy-loaded when sampling a mini-batch. The slide queue $W^{(\text{queue})}$ is not used during the first epoch to ensure it is filled, and the prototypes are also frozen during these epochs. The model training was monitored with Weight & Biases. The hyperparameters were fined-tuned with the sweep option from Weight & Biases on a heuristic, the product of the FIDE score and the Shannon entropy of the spatial domains distribution. The results of a hyperparameter search are described in Supplementary Notes and illustrated in Supplementary Figs. 7, 8 and 14. Novae was trained on an Nvidia HGX A100 GPU for 24 h.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The MERSCOPE datasets are available online at https://vizgen.com/data-release-program/. The Xenium datasets are available online at

https://www.10xgenomics.com/datasets. The CosMX datasets are available online at https://nanostring.com/products/cosmx-spatial-molecular-imager/ffpe-dataset/. The scripts to download these datasets are available in the repository of Novae at https://github.com/MICS-Lab/novae/tree/main/data, and Supplementary Data 1 details which samples are used for the different study cases. Every dataset used in the paper, including the toy dataset and the MGC dataset, is also available on the Hugging Face Hub at https://huggingface.co/datasets/MICS-Lab/novae and can be directly downloaded via the Novae package API.

Code availability

The code developed in this article is available as an open-source Python package, accessible on GitHub at https://github.com/MICS-Lab/novae or via Zenodo at https://doi.org/10.5281/zenodo.16785552 (ref. 58). Additionally, scripts with usage examples are accessible on Code Ocean at https://doi.org/10.24433/CO.8708559.v1 (ref. 59). The code used to run the benchmark is available at https://github.com/MICS-Lab/novae_benchmark.

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Acknowledgements

This work is supported by IHU Prism funded by the France 2030 programme and the French National Research Agency (ANR) under grant number ANR-23-IAHU-0002 (to F.A. and P.H.C.), as well as the Fondation ARC Pour la Recherche sur le Cancer.

Author contributions

Q.B., F.A. and P.-H.C. defined the objectives of the project. Q.B. defined the methodological approach, implemented Novae and wrote the paper. H.B. performed the computational benchmark between Novae and the existing state-of-the-art models, gave ideas for the methodological approach and contributed to writing the paper. N.B. validated the biological results of the model and wrote the section related to these findings. F.A and F.G. validated the biological results of the model and the clinical relevance of the study. G.G. provided the MGC dataset and annotated the H&E domains for the multimodal lung showcase. K.M. performed the Novae biological analysis on both the MGC and lung dataset. P.-H.C. supervised Q.B., concentrating on the methodological aspects of the work and paper development. F.A. and P.-H.C. acquired the grant to fund the project. All authors validated the paper and contributed to improving it.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41592-025-02899-6.

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Peer review information *Nature Methods* thanks Jesper Tegner, Ka-Chun Wong for their contribution to the peer review of this work. Peer reviewer reports are available. Primary Handling Editor: Lin Tang, in collaboration with the *Nature Methods* team.

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Last updated by author(s): Aug 9, 2025

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Policy information about availability of computer code

Data collection	Public datasets were used for this manuscript, with only one dataset being non-public, but coming from another paper of our institute (see reference in the manuscript).
Data analysis	The code developed in this article is available as an open-source Python package called "novae" accessible on PyPI and Github at https://github.com/MICS-Lab/novae. Its version 0.2.2 has been used for this manuscript. The code used to run the benchmark is available at https://github.com/MICS-Lab/novae_benchmark. The main dependencies of novae includes scanpy (we used version 1.11.1), torch (2.7.0), and torch-geometric (2.6.1).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The MERSCOPE datasets are available online at https://vizgen.com/data-release-program/.
The Xenium datasets are available online at https://www.10xgenomics.com/datasets.
The CosMX datasets are available online at https://nanostring.com/products/cosmx-spatial-molecular-imager/ffpe-dataset/. The scripts to download all the datasets are also available in the repository of Novae at https://github.com/MICS-Lab/novae/tree/main/data. The toy dataset and the MGC dataset used in this manuscript is available on Hugging Face at https://huggingface.co/datasets/MICS-Lab/novae/tree/main and can be directly downloaded via the Novae package API.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	This information has not been collected, as the research is not involving new human participants
Reporting on race, ethnicity, or other socially relevant groupings	This information has not been collected, as the research is not involving new human participants
Population characteristics	This information has not been collected, as the research is not involving new human participants
Recruitment	No patient has been recruited
Ethics oversight	There is no study protocol

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We tried to collect as many public samples as possible, i.e. 78 slides in total. We also made public an entire dataset from a previous publication of our institute. The sample size for the latter dataset was therefore determined by the previous publication, which was referenced in our article.
Data exclusions	No data has been excluded from the analysis. For each technology, we have chosen the largest public dataset available.
Replication	The code developed is fully reproducible.
Randomization	Randomisation was not needed in the study as there were no biases from the start of the experiment. Nor did we have study groups to compare in our analysis.
Blinding	During the selection of the public datasets, we have only chosen the largest public datasets available, without first looking at the data.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Not involved in the study
Novel plant genotypes	Not involved in the study
Authentication	Not involved in the study